has one sign change. By our Descartes rule, the number of positive zeros of the polynomial $P(x)$ cannot be more than 2; the number of negative zeros of the polynomial $P(x)$ cannot be more than 1. Clearly 1 and 2 are positive zeros, and -1 is the negative zero for the polynomial, $x^5 - 2x^4 - x + 2$, and hence the bounds 2 for positive zeros and the bound 1 for negative zeros are attained. We note that i and $-i$ are neither positive nor negative.

We know $(x+2)(x+3)(x+i)(x-i)$ is a polynomial with roots $-2, -3, -i, i$. The polynomial, say $P(x)$, in the customary form is $x^4 + 5x^3 + 7x^2 + 5x + 6$.

This polynomial $P(x)$ has no sign change and $P(-x) = x^4 - 5x^3 + 7x^2 - 5x + 6$ has 4 sign changes. By Descartes rule, the polynomial $P(x)$ cannot have more than 0 positive zeros and the number of negative zeros of the polynomial $P(x)$ cannot be more than 4.

As another example, we consider the polynomial.

$$x^n - {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} - {}^nC_3 x^{n-3} + \cdots + (-1)^{n-1}\, {}^nC_{(n-1)} x + (-1)^n .$$

This is the expansion of $(x-1)^n$. This polynomial has n changes in coefficients and $P(-x)$ has no change of sign in coefficients. This shows that the number of positive zeros of the polynomial cannot be more than n and the number of negative zeros of the polynomial cannot be more than 0. The statement on negative zeros gives a very useful information that the polynomial has no negative zeros. But the statement on positive zeros gives no good information about the positive zeros, though there are exactly n positive zeros; in fact, it is well-known that for a polynomial of degree n, the number of zeros cannot be more than n and hence the number of positive zeros cannot be more than n.

3.9.2 (b) Bounds for the number of Imaginary (Nonreal Complex)roots

Using the Descartes rule, we can compute a lower bound for the number of imaginary roots. Let m denote the number of sign changes in coefficients of $P(x)$ of degree n; let k denote the number of sign changes in coefficients of $P(-x)$. Then there are at least $n-(m+k)$ imaginary roots for the polynomial $P(x)$. Using the other conclusion of the rule, namely, the difference between the number of roots and the corresponding sign changes is even, we can sharpen the bounds in particular cases.

Example 3.30

Show that the polynomial $9x^9 + 2x^5 - x^4 - 7x^2 + 2$ has at least six imaginary roots.

Solution

Clearly there are 2 sign changes for the given polynomial $P(x)$ and hence number of positive roots of $P(x)$ cannot be more than two. Further, as $P(-x) = -9x^9 - 2x^5 - x^4 - 7x^2 + 2$, there is one sign change for $P(-x)$ and hence the number of negative roots cannot be more than one. Clearly 0 is not a root. So maximum number of real roots is 3 and hence there are atleast six imaginary roots. ■

Remark From the above discussion we note that the Descartes rule gives only upper bounds for the number of positive roots and number of negative roots; the Descartes rule neither gives the exact number of positive roots nor the exact number of negative roots. But we can find the exact number of positive, negative and nonreal roots in certain cases. Also, it does not give any method to find the roots.

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